Biomechanics-Driven Optimization of Exoskeleton Configuration and Control for Knee Contact Force Reduction in Inclined Walking

Wei Jin, Jiangtian Li, Zilu Wang, Fengchen Liu, Zhihao Zhou, and Qining Wang

Abstract— **Knee osteoarthritis is a prevalent musculoskeletal disorder characterized by excessive internal joint loading, which is further exacerbated during demanding locomotor tasks such as inclined walking. While lower-limb exoskeletons offer potential for assistance, most existing designs are developed with generic objectives that do not explicitly target the reduction of knee contact force (KCF), a direct indicator of joint loading. This study proposes a biomechanics-driven optimization framework to identify the optimal configuration and control parameters of a knee exoskeleton for KCF reduction during inclined walking. The framework integrates human-exoskeleton interaction modeling, subject-specific musculoskeletal simulation, encompassing inverse dynamics, static optimization, and joint reaction analysis, and particle swarm optimization. Simulation results demonstrate that the optimized parameters achieved a decrease in mean KCF from 17.69 N/kg (no assistance) to 17.04 N/kg (pre-optimized assistance) and further to 16.60 N/kg (optimized assistance). Experimental validation (15°, 0.6 m/s) confirmed these findings, showing that optimized assistance reduced mean KCF from 17.58 N/kg to 16.45 N/kg. Furthermore, RMS muscle activations for the quadriceps and hamstrings were reduced by 16.2% and 22.0%. These findings demonstrate the framework's effectiveness in customizing exoskeleton parameters to alleviate internal joint loading, offering a principled approach for clinical gait assistance.**

I. INTRODUCTION

Knee osteoarthritis is one of the most prevalent musculoskeletal disorders worldwide and a leading cause of pain, mobility impairment, and reduced quality of life among older adults [1]. As the disease progresses, patients experience increasing difficulty in daily locomotor tasks, particularly those involving elevated mechanical demands such as uphill walking and stair ascent [2], [3]. These activities substantially amplify joint loading at the knee, thereby accelerating cartilage degeneration and exacerbating pain symptoms [4]. Consequently, mitigating excessive knee joint loading during locomotion has become a central objective in both clinical intervention and assistive device design for individuals with knee osteoarthritis [5]–[9].

From a biomechanical perspective, knee contact force (KCF) provides a direct and physiologically meaningful measure of internal joint loading [10]. Elevated KCF is strongly associated with cartilage wear, disease progression, and symptom severity in patients with knee osteoarthritis [11]. Compared to external metrics such as ground reaction forces or net joint moments, KCF more explicitly reflects the combined effects of muscle forces, external loads, and joint kinematics on the tibiofemoral joint [3]. Notably, inclined walking induces a marked increase in KCF due to greater knee extension demands during stance, making it a particularly relevant and challenging scenario for load-reduction strategies.

Lower-limb exoskeletons have emerged as a promising approach for assisting and alleviating musculoskeletal burden. Previous studies have demonstrated the potential of knee exoskeletons to reduce metabolic cost, augment joint torque, and redistribute muscle activity during walking [12]–[14]. However, most existing designs and control strategies are developed with generic performance objectives, such as minimizing metabolic energy expenditure or providing predefined assistive torques [8], [12], [15]. While these approaches may indirectly influence joint loading, they do not explicitly target KCF reduction. As a result, a general-purpose knee exoskeleton configuration may not be optimal, or even effective, for mitigating joint contact forces, particularly under demanding conditions such as uphill walking.

The effectiveness of an exoskeleton in reducing KCF depends not only on the magnitude and timing of the assistive force, but also critically on the geometric configuration of the human-exoskeleton interface. Parameters such as attachment locations directly influence the moment arm of the assistive force and its interaction with the musculoskeletal system. These configuration and control parameters are highly coupled, and their combined effects on joint loading are difficult to predict through intuition or empirical tuning alone. Musculoskeletal simulation offers a powerful tool for estimating internal joint forces and muscle activations, enabling systematic evaluation of biomechanical outcomes that are otherwise inaccessible *in vivo* [16], [17]. When integrated with numerical optimization, such simulations provide a principled framework for identifying exoskeleton designs that are tailored to specific biomechanical objectives.

In this study, we propose a biomechanics-driven optimiza-

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Wei Jin and Zilu Wang are with the School of Advanced Manufacturing and Robotics, Peking University, Beijing, China.

Jiangtian Li is with the School of Mechanics and Engineering Science, Peking University, Beijing, China.

Fengchen Liu is with the School of Biological Science and Medical Engineering, Beihang University, Beijing, China, and also with the School of Rehabilitation Sciences and Engineering, University of Health and Rehabilitation Sciences, Qingdao, China.

Zhihao Zhou is with the Institute for Artificial Intelligence, Peking University, Beijing 100871, China, and also with the Beijing Engineering Research Center of Intelligent Rehabilitation Engineering, Beijing, China.

Qining Wang is with the School of Advanced Manufacturing and Robotics, and the Institute for Artificial Intelligence, Peking University, Beijing, China, and also with the School of Rehabilitation Sciences and Engineering, University of Health and Rehabilitation Sciences, Qingdao, China (qiningwang@pku.edu.cn).

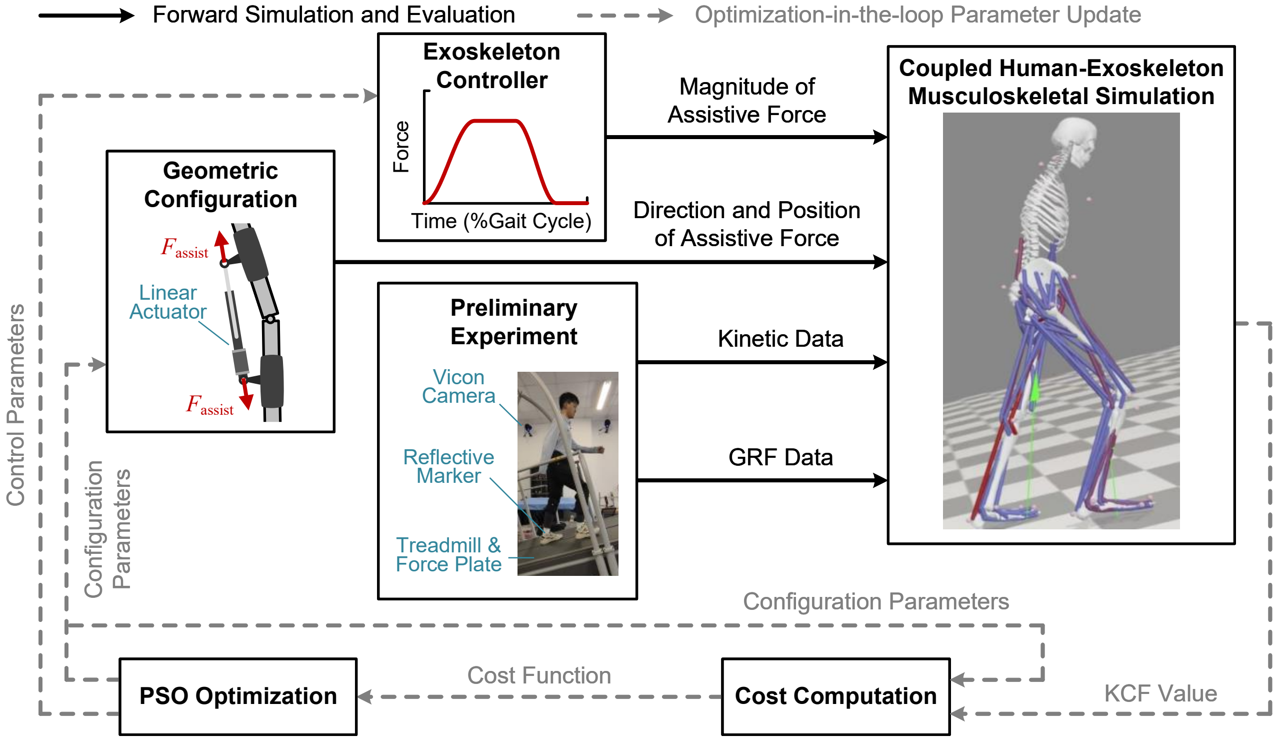

Fig. 1. Overview of the biomechanics-driven optimization framework for knee exoskeleton parameter identification. Subject-specific kinematic and GRF data from a preliminary inclined walking experiment are used as fixed inputs to a coupled human-exoskeleton musculoskeletal simulation, which encompasses *Inverse Dynamics*, *Static Optimization*, and *Joint Reaction Analysis* to evaluate KCF. The knee exoskeleton is modeled as a parameterized assistive system, in which configuration and control parameters jointly define the magnitude, direction, and point of application of the assistive force. The PSO algorithm iteratively updates the exoskeleton parameters by minimizing a composite objective that accounts for both KCF reduction and wearability.

tion framework for knee exoskeleton parameter identification, with the explicit goal of reducing KCF during inclined walking. The framework integrates human-exoskeleton interaction modeling, subject-specific musculoskeletal simulation, and particle swarm optimization (PSO) to jointly optimize both configuration parameters and control strategy parameters of a knee exoskeleton. A composite objective function is formulated to prioritize KCF reduction while accounting for wearability-related considerations. The optimized parameters are subsequently implemented on a physical knee exoskeleton system and evaluated experimentally. By focusing on KCF as the primary biomechanical target, this work demonstrates a systematic approach for customizing exoskeleton design and control to address joint loading in challenging locomotor tasks.

II. Methods

A. Overview of the Biomechanics-Driven Optimization Framework

This study proposes a biomechanics-driven optimization framework that integrates human-exoskeleton interaction modeling, musculoskeletal dynamic simulation, and optimization-in-the-loop parameter updating to identify exoskeleton configuration and control strategy for reducing KCF during inclined walking. The overall workflow of the proposed framework is illustrated in Fig. 1.

First, the subject-specific kinematic and ground reaction force (GRF) data are collected in a preliminary treadmill walking experiment. These experimental data serve as fixed biomechanical inputs to the musculoskeletal simulation and remain unchanged throughout the optimization process. The knee exoskeleton is modeled as a parameterized assistive system, in which configuration-related parameters determine the direction and point of application of the assistive force, while control-related parameters define its magnitude and temporal profile over the gait cycle. For a given set of exoskeleton parameters, the corresponding assistive forces are computed and applied to a coupled human-exoskeleton musculoskeletal model.

A forward simulation and evaluation pipeline consisting of *Inverse Dynamics*, *Static Optimization*, and *Joint Reaction Analysis* is then executed to estimate the resulting KCF. The mean KCF extracted from the simulation, together with wearability-related metrics, is used to compute the composite cost function. The PSO algorithm operates in an optimization-in-the-loop manner, iteratively updating the exoskeleton parameters based on the evaluated cost. At each iteration, a population of candidate parameter sets is generated, evaluated through the musculoskeletal simulation pipeline, and refined according to the optimization algorithm. This iterative process continues until predefined convergence criteria are met, yielding an optimized set of exoskeleton parameters. The following subsections provide detailed de-

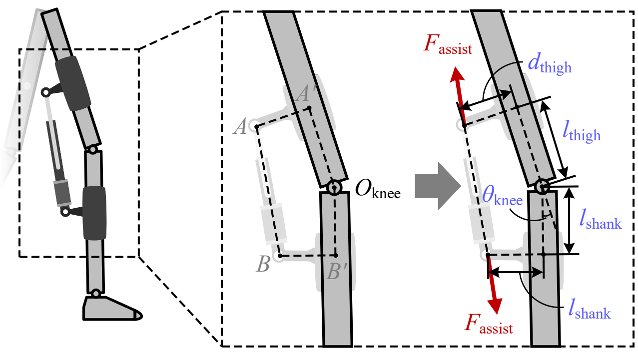

Fig. 2. Geometric model of the human-exoskeleton interaction at the knee joint. The assistive force F_{assist} is applied to the thigh and shank through posterior cuffs. The configuration is defined by the lateral offsets d_{thigh}, d_{shank}, and the longitudinal distances l_{thigh}, l_{shank} with respect to the knee joint center θ_{knee}.

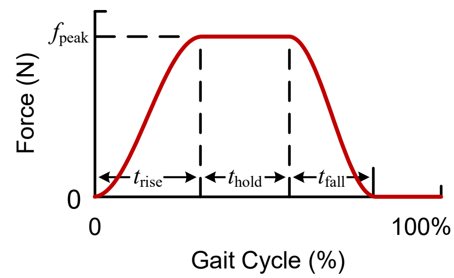

Fig. 3. Parameterized temporal profile of the assistive knee extension force over a normalized gait cycle. The force magnitude is defined by a peak value f_{peak} and three timing parameters: rise time t_{rise}, hold time t_{hold}, and fall time t_{fall}, expressed as percentages of the gait cycle.

scriptions of the interaction modeling, objective formulation, simulation-based evaluation, and optimization procedure.

B. Human-Exoskeleton Interaction Modeling

A rigid-soft hybrid knee exoskeleton, previously introduced in detail in [18], is employed in this study and mechanically coupled to the human lower limb. As illustrated in Fig. 2, a linear actuator is positioned posterior to the knee joint and generates bidirectional assistive forces, denoted as F_{assist} on the posterior aspects of the thigh and shank through compliant cuffs. The two ends of the actuator are hinged to the thigh cuff and shank cuff, respectively, allowing relative rotation during knee flexion and extension.

The geometric configuration of the human-exoskeleton interface is parameterized using a set of distances defined with respect to the anatomical reference frames of the thigh and shank. Specifically, the lateral distance from the upper actuator hinge point (A) to the central axis of the thigh segment is denoted as d_{thigh}, while the corresponding distance from the lower hinge point (B) to the central axis of the shank segment is denoted as d_{shank}. In addition, the longitudinal distances from the centers of thigh and shank cuffs (A' and B') to the knee joint center (O_{knee}) are denoted as l_{thigh} and l_{shank}, respectively. Together, these parameters define the spatial configuration and positioning of the exoskeleton relative to the human limb.

The temporal profile of the assistive knee extension force over a normalized gait cycle is illustrated in Fig. 3. A gait cycle is defined as the interval between two consecutive heel strikes of the same foot [19]. The assistive force profile is parameterized using four variables: rise time t_{rise}, hold time t_{hold}, fall time t_{fall}, and peak force f_{peak}. The three temporal parameters are expressed as percentages of the gait cycle (%GC). To ensure smooth force transitions and avoid discontinuities in the applied interaction forces, the rising and falling phases are implemented using sinusoidal interpolation, which guarantees first-derivative continuity.

Based on the above human–exoskeleton interaction model, and assuming that the peak force f_{peak} is fixed, the configuration and control strategy of the knee exoskeleton are fully characterized by seven parameters: the attachment-related parameters d_{thigh}, d_{shank}, l_{thigh}, l_{shank} in meters, and the timing-related control parameters t_{rise}, t_{hold}, t_{fall} in %GC. These parameters jointly determine the magnitude, direction, and point of application of the assistive force F_{assist} throughout the gait cycle. They are therefore selected as optimization variables in the proposed framework, as defined in the following section.

C. Composite Objective Formulation and Optimization Variables

The optimization objective is formulated to account for biomechanical effectiveness and wearability considerations simultaneously. Specifically, the primary objective is to minimize the KCF during inclined walking ($f_{\text{KCF,mean}}$), which is strongly associated with joint loading severity in osteoarthritis patients [20]. In addition, the lateral attachment offsets of the exoskeleton on the thigh (d_{thigh}) and shank (d_{shank}) are included in the objective to penalize excessive protrusion and ensure wearable and mechanically stable configurations.

The overall optimization objective $J(\mathbf{p})$ is formulated as a weighted sum of normalized terms:

$$J(\mathbf{p}) = w_1 \hat{f}_{\text{KCF,mean}} + w_2 \hat{d}_{\text{thigh}} + w_3 \hat{d}_{\text{shank}}, \qquad (1)$$

where the normalized components ($\hat{f}_{\text{KCF,mean}}$, $\hat{d}_{\text{thigh}}$, and $\hat{d}_{\text{shank}}$) are defined as

$$\begin{aligned} \hat{f}_{\text{KCF,mean}} &= f_{\text{KCF,mean}}/G, \\ \hat{d}_{\text{thigh}} &= d_{\text{thigh}}/r_{\text{thigh}}, \\ \hat{d}_{\text{shank}} &= d_{\text{shank}}/r_{\text{shank}}. \end{aligned}$$

In (1), $\mathbf{p} = [d_{\text{thigh}}, d_{\text{shank}}, l_{\text{thigh}}, l_{\text{shank}}, t_{\text{rise}}, t_{\text{hold}}, t_{\text{fall}}]^{\mathsf{T}}$ denotes the hybrid parameter vector encompassing both configuration and control strategy of the exoskeleton. The normalization factors include the subject's body weight G and the respective radii of the thigh (r_{thigh}) and shank (r_{shank}) at the cuff locations. The non-negative weighting factors w_i satisfy $\sum w_i = 1$, and are tuned to prioritize KCF reduction while penalizing excessive lateral offsets. By integrating these competing objectives into a single cost function, the proposed framework enables systematic exploration of configurations that are both biomechanically

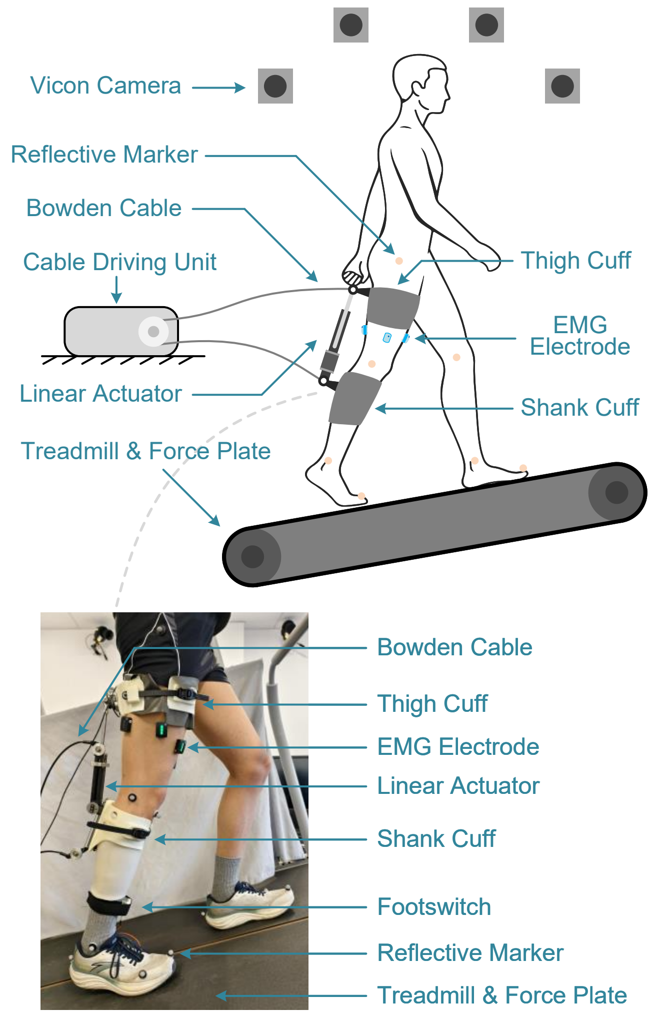

Fig. 4. Experimental setup and knee exoskeleton implementation for inclined walking. A schematic illustration (top) and a photograph (bottom) show the knee exoskeleton, sensing components, and instrumented treadmill used for inclined walking experiments. Kinematic, GRF, and EMG data were synchronously collected during walking trials.

effective and wearable. The feasible search space for the optimization variables is defined as: $d_{\text{thigh}} \in [0.08, 0.15]$ m, $d_{\text{shank}} \in [0.06, 0.15]$ m, $l_{\text{thigh}} \in [0.05, 0.25]$ m, $l_{\text{shank}} \in [0.14, 0.24]$ m, $t_{\text{rise}} \in [5, 15]$ %GC, $t_{\text{hold}} \in [15, 50]$ %GC, $t_{\text{fall}} \in [5, 30]$ %GC.

D. Musculoskeletal Simulation-Based Evaluation

1) Computational Workflow for KCF Evaluation: To evaluate the impact of exoskeleton parameters on joint loading, we developed a simulation pipeline that integrates human-exoskeleton interaction with musculoskeletal modeling. The joint kinetics are computed via *Inverse Dynamics*, which takes joint kinematics, GRFs, and the exoskeleton's assistive forces as inputs. Specifically, the assistive forces, modeled based on the interaction dynamics described in Section II-B, are applied as external point loads to the thigh and shank segments at the mounting locations.

Following *Inverse Dynamics*, *Static Optimization* is employed to resolve the muscle redundancy problem and estimate the muscle forces. Finally, the *Joint Reaction Analysis* function is utilized to quantify the resulting KCF. This iterative pipeline allows for the systematic evaluation of how different parameter combinations modulate both muscle recruitment patterns and knee joint loading.

2) Preliminary Data Acquisition and Simulation Setup: The framework is implemented in MATLAB (R2024b, The MathWorks, Natick, MA, USA) with OpenSim (version 4.5) API. To provide the necessary biomechanical inputs for the optimization, a preliminary experiment was conducted on a healthy male subject (184 cm, 70.6 kg; thigh radius at the cuff [r_{thigh}]: 7.96 cm, shank radius at the cuff [r_{shank}]: 5.57 cm). The subject performed walking trials on an instrumented split-belt treadmill (Bertec Corp., Columbus, OH, USA) at 0.6 m/s with a 15° incline (Fig. 4). 26 reflective markers were placed on the subject's anatomical landmarks based on an improved set from [17]. A 12-camera motion capture system (Vicon, Oxford, UK) was employed to track the marker positions at 100 Hz, and GRF data were recorded at 1 KHz via the treadmill equipped with force plates. A force-sensing resistor is integrated into the heel of the insole as a footswitch to detect heel strike events by triggering a binary gait-phase signal once a predefined pressure threshold is exceeded. A single representative gait cycle was selected from the collected trial and used as the fixed biomechanical input throughout the optimization process.

The OpenSim Gait2392 model was utilized as the biomechanical template model for the musculoskeletal simulation [21]. The scaling tool first customized the model's anthropometry to match the subject based on the marker positions during a static motion capture trial. Subsequently, *Inverse Kinematics* was performed to derive the joint angles by minimizing the error between the experimental and virtual marker trajectories.

3) Optimization Parameters and Assumptions: For the composite objective function (Eq. (1)), the weighting factors were empirically assigned as $w_1 = 0.8$, $w_2 = 0.1$, $w_3 = 0.1$. The dominant weight on $\hat{f}_{\text{KCF,mean}}$ reflects the primary biomechanical objective, while the smaller weights on d_{thigh} and d_{shank} serve as regularization to prevent excessively large lateral protrusion that would compromise attachment stability and force transmission. It is important to note that the unilateral knee exoskeleton used in this study weighs only 1.2 kg [18], and thus its impact on gait dynamics is assumed to be negligible. Consequently, the physical exoskeleton's mass or inertial properties are not modeled within the musculoskeletal model. The force transmission losses are not considered to reduce computational complexity during the iterative optimization process.

E. PSO-based Optimization Procedure

The proposed optimization is performed in a simulation-in-the-loop manner, where each candidate parameter vector is evaluated through the musculoskeletal simulation pipeline described in Section II-D. Specifically, for a given parameter vector $\mathbf{p}$, the corresponding human-exoskeleton interaction forces are computed and applied to the OpenSim model, followed by *Inverse Dynamics*, *Static Optimization*, and *Joint Reaction Analysis* to obtain the $f_{\mathrm{KCF,mean}}$. The resulting biomechanical outcome, together with the wearability-related parameters, is then used to evaluate the composite objective function defined in Eq. (1). This closed-loop evaluation enables systematic exploration of the design space while explicitly accounting for subject-specific biomechanics.

PSO is employed to update the candidate solutions across iterations [22]–[24]. Owing to its derivative-free nature and strong global search capability, PSO is well-suited for the present problem, which involves a non-linear, high-dimensional objective function evaluated through computationally intensive musculoskeletal simulations [25]. The PSO algorithm is initialized with a population of n particles, each representing a candidate parameter vector $\mathbf{p}^k$. At each iteration, the objective function $J(\mathbf{p^k})$ is evaluated for all particles via the simulation pipeline. For each particle, the local best position encountered so far, denoted as $\mathbf{p}^k_{\mathrm{pb}}$, is recorded, along with the position $\mathbf{p}^k_{\mathrm{nb}}$ that yields the minimum objective value among the neighborhood. The velocity of the particle k at iteration i is updated according to

$$\mathbf{v}_i^k = W_i\mathbf{v}_{i-1}^k + \gamma_1 u_1\left(\mathbf{p}^k_{\mathrm{pb}} - \mathbf{p}_i^k\right) + \gamma_2 u_2\left(\mathbf{p}^k_{\mathrm{nb}} - \mathbf{p}_i^k\right), \quad (2)$$

where W is the inertia weight, γ_1 and γ_2 are the cognitive and social learning coefficients, u_1 and u_2 are independent random values uniformly distributed in $[0, 1]$. The particle position is then updated as

$$\mathbf{p}_{i+1}^k = \mathbf{p}_i^k + \mathbf{v}_i^k. \quad (3)$$

To mitigate premature convergence and enhance global exploration, a local neighborhood topology is adopted, where the neighborhood size is initially set to a small fraction of the population and is adaptively increased if no improvement in the global best solution is observed over successive iterations [26]. Boundary constraints are enforced after each update to ensure that all particles remain within the predefined feasible design space.

The optimization process terminates when either the maximum number of iterations N is reached or when the relative improvement of the global best objective over N_c iterations falls below a predefined tolerance. In this study, the population size is set to $n = 12$ to balance sward diversity against the per-iteration computational cost of the simulation; $N = 200 \cdot \dim(\mathbf{p}) = 1400$, following the default setting of MATLAB's `particleswarm` solver; N_c is set to 50, which provides a balance between convergence reliability and computational cost.

III. Experimental Validation and Data Analysis

A. Exoskeleton Hardware Implementation

A unilateral knee exoskeleton with a rigid-soft hybrid architecture, previously introduced in [18], was employed for experimental validation (Fig. 4). The device consists of a wearable module that mechanically couples to the thigh and shank via compliant cuffs, and a remote cable-driven unit that powers the actuator via a bidirectional cable drive. By varying the actuator length, assistive forces are transmitted between the thigh and shank, enabling bidirectional assistance during knee flexion and extension.

The exoskeleton design allows flexible adjustment of the attachment configuration, including both the longitudinal placement of the cuffs along the limb segments and the lateral offset of the actuator relative to the segment centerlines. These adjustable features directly correspond to the optimized geometric parameters (d_{thigh}, d_{shank}, l_{thigh}, and l_{shank}) identified in Section II-B, thereby enabling the physical realization of the optimization results.

The wearable component is lightweight (1.2 kg), and its mass is primarily distributed along the limb segments. As a result, the exoskeleton is assumed to introduce negligible inertial effects on gait dynamics, consistent with the modeling assumptions adopted in the musculoskeletal simulations.

B. Experimental Protocol

A single healthy male subject participated in the experiment. The same subject as in the preliminary data acquisition was intentionally selected to ensure consistency between the subject-specific musculoskeletal model and the optimization results. Anthropometric information, walking conditions, and experimental setup were identical to the preliminary experiment described in Section II-D. Briefly, the subject walked on an instrumented split-belt treadmill for 2 minutes at a speed of 0.6 m/s with a 15° incline, while kinematic and kinetic data were collected using a motion capture system and embedded force plates. Marker placement and data preprocessing procedures (including model scaling and *Inverse Kinematics*) were kept unchanged to ensure methodological consistency.

Muscle activation patterns were recorded using a 16-channel wireless surface electromyography (EMG) system (Trigno Wireless EMG System, Delsys Inc., Natick, MA, USA) at a sampling rate of 2000 Hz. EMG signals were collected from five primary muscles surrounding the knee joint of the assisted leg: rectus femoris, vastus lateralis, vastus medialis, semitendinosus, and biceps femoris.

Following the optimization procedure described in Section II-E, the exoskeleton was physically configured according to the optimized geometric parameters, and the assistive force timing parameters were implemented in the control system. The subject then performed walking trials under the optimized assistance condition as well as an unassisted baseline condition for comparison. Given the single-subject design, the experimental results are intended to provide subject-specific validation rather than population-level statistical inference.

C. Data Analysis

Raw EMG signals were band-pass filtered (20—450 Hz), full-wave rectified, and low-pass filtered (5 Hz) to generate linear envelopes, which were then normalized to the unassisted peak values. The activation envelopes of rectus femoris, vastus lateralis, and vastus medialis were combined to represent the quadriceps muscle group, while the envelopes of semitendinosus and biceps femoris were combined to represent the hamstrings muscle group. Knee torque was computed through the *Inverse Dynamics* function. KCF was estimated through the *Joint Reaction Analysis* function, as described in Section II-D, with mean values extracted as primary metrics to quantify the load-reduction performance. The magnitudes of knee torque and KCF values were normalized to the model's body weight.

IV. Results and Discussion

A. Optimization Outcomes and Simulation Analysis

The optimization process exhibited stable convergence, indicating a well-posed composite objective formulation. Starting from the initial parameter set $\mathbf{p}_0 = [d_{\text{thigh}}, d_{\text{shank}}, l_{\text{thigh}}, l_{\text{shank}}, t_{\text{rise}}, t_{\text{hold}}, t_{\text{fall}}]^{\mathsf{T}} = [0.10, 0.12, 0.10, 0.15, 10, 20, 10]$, the cost function value decreased from $J(\mathbf{p}_0) = 1.6887$ to $J(\mathbf{p}_{66}) = 1.5562$ after 66 iterations, corresponding to an overall reduction of approximately 7.8%. The stopping criterion based on the maximum number of non-improving iterations (N_c) was reached. As shown in Fig. 5(a), most of the improvement occurred during the early stage of optimization, followed by gradual convergence toward a stable solution.

The optimized set $\mathbf{p}_{66} = [0.080, 0.060, 0.129, 0.161, 5.047, 30.133, 5.450]^{\mathsf{T}}$ reveals coordinated adjustments in both exoskeleton configuration and control timing. Compared to the initial design, the optimized solution favors shorter lateral offsets at both the thigh and shank attachments, despite the associated reduction in moment arm. This configuration improves attachment stability and force transmission robustness, reducing relative motion between the exoskeleton and the limb during dynamic gait. In contrast, the effective lever arms along the segments (l_{thigh} and l_{shank}) are increased, compensating for the reduced lateral offsets by enhancing torque generation efficiency through a longer longitudinal moment contribution. From a control perspective, the optimized assistance profile exhibits faster rise and fall times combined with a substantially prolonged hold phase. This timing strategy concentrates assistance during the mechanically relevant portion of stance while avoiding prolonged transitions that could interfere with natural joint dynamics. Together, these configurations and control adjustments enable more effective unloading of the knee joint while maintaining stable and realizable assistance delivery.

As illustrated in Fig. 5(b), peak knee torque decreased from 0.543 Nm/kg in the no-assistance condition to 0.428 Nm/kg with pre-optimized assistance, and further to 0.378 Nm/kg after optimization. Muscle activation patterns showed a similar trend. The RMS activation of the quadriceps was reduced from 0.120 (no assistance) to 0.102 (pre-optimized) and 0.095 (optimized), while hamstrings RMS activation remained comparable across conditions (0.137, 0.140, and 0.137, respectively), indicating that the optimized assistance primarily alleviated extensor demand without increasing antagonist co-contraction (Fig. 5(c)).

Regarding joint loading, the optimized assistance achieved a reduction in KCF during early stance, where peak KCF

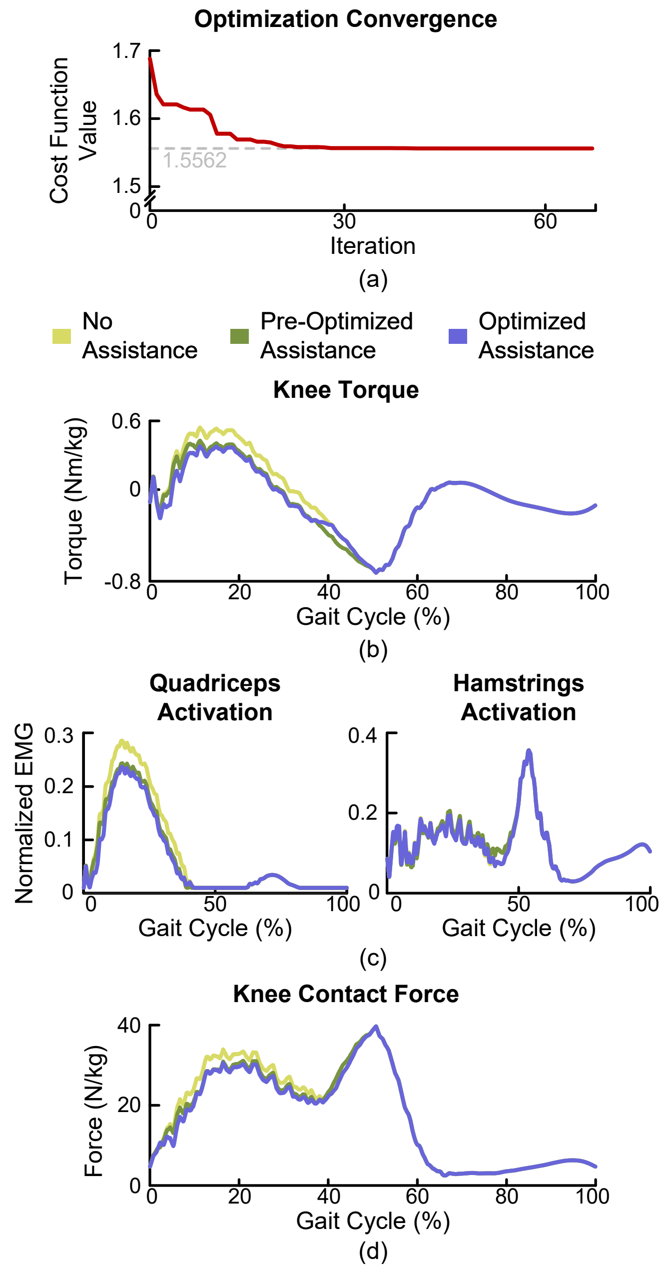

Fig. 5. Simulation results of optimization convergence and knee biomechanical responses during inclined walking. (a) Convergence of the composite cost function over optimization iterations. (b) Knee torque, (c) quadriceps and hamstrings activations, and (d) KCF, all normalized to the gait cycle. Results in (b)–(d) are shown for the no-assistance (yellow), pre-optimized assistance (green), and optimized assistance (blue) conditions, derived from the preliminary walking trial.

decreased from 34.00 N/kg (no assistance) to 30.97 N/kg (pre-optimized) and 30.61 N/kg (optimized). Moreover, the mean KCF over the gait cycle was reduced from 17.69 N/kg to 17.04 N/kg and further to 16.60 N/kg, respectively (Fig. 5(d)). These results demonstrate that the proposed optimization framework effectively improves the KCF-oriented performance of the exoskeleton beyond an intuitive, non-optimized configuration.

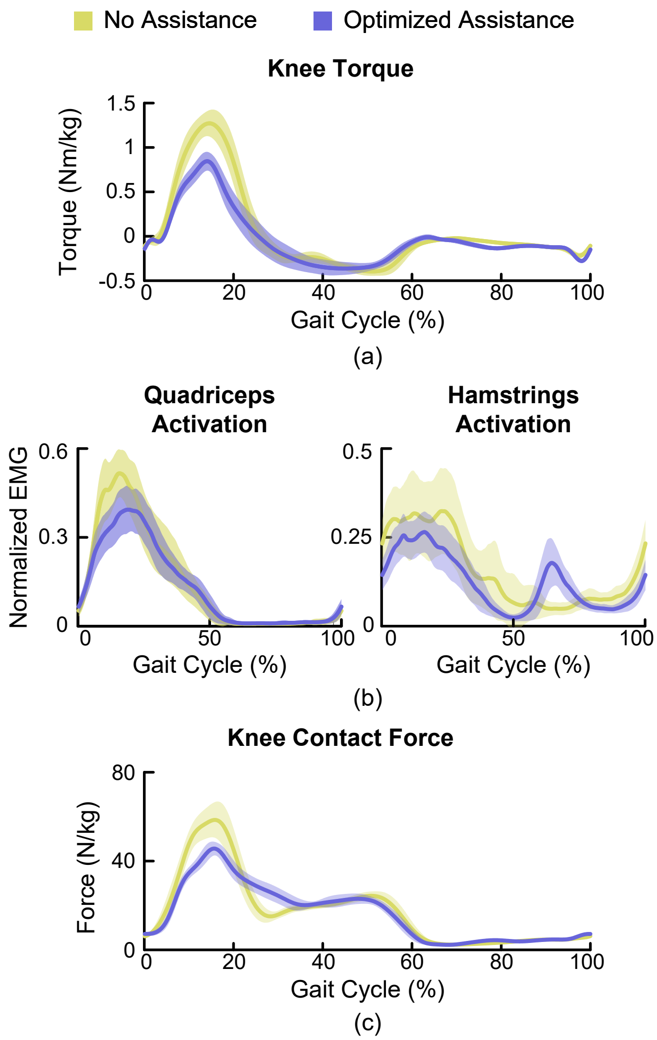

Fig. 6. Experimental comparison of knee biomechanics between the no-assistance and optimized assistance conditions during inclined walking. (a) Knee torque, (b) quadriceps and hamstrings activation, and (c) KCF over a normalized gait cycle. Solid lines with shaded areas represent the mean ± 1 standard deviation across gait cycles extracted from the 2-minute walking trial for the no-assistance (yellow) and optimized assistance (blue) conditions.

B. Experimental Validation of Optimized Parameters

Following the simulation-based optimization and analysis, experimental validation was conducted to examine whether the optimized exoskeleton parameters yield consistent biomechanical benefits during human walking. Fig. 6 presents the experimentally measured knee joint biomechanics under the no-assistance and optimized assistance conditions during inclined walking. Consistent with the simulation results, the optimized assistance led to a clear reduction in knee joint loading and muscle demand, while preserving the overall temporal structure of gait. As shown in Fig. 6(a), the peak knee torque during early stance decreased from 1.27 Nm/kg in the no-assistance condition to 0.85 Nm/kg with optimized assistance, indicating a substantial reduction in biological knee extensor demand during the loading response state of inclined walking.

Correspondingly, quadriceps activation was reduced in both peak (0.515 to 0.393) and RMS values (0.228 to 0.191), reflecting a sustained unloading effect throughout the gait cycle (Fig. 6(b)). Hamstrings activation also decreased in peak (0.324 to 0.265) and RMS (0.186 to 0.145) metrics, suggesting that the optimized assistance did not induce excessive antagonist co-contraction and maintained balanced muscle coordination around the knee joint (Fig. 6(b)). These findings align with the simulation outcomes, in which reductions in muscle activation accompanied decreases in knee joint loading.

KCF also exhibited reductions in both peak and mean values (Fig. 6(c)) under the optimized assistance condition. Peak KCF decreased from 58.55 N/kg to 45.63 N/kg, while mean KCF decreased from 17.58 N/kg to 16.45 N/kg. This outcome directly reflects the optimization objective, which prioritized KCF reduction through musculoskeletal simulation-based evaluation. Although the magnitude of KCF reduction differs from that observed in simulation, the experimental results confirm the same overall trend: optimized assistance effectively reduces internal knee joint loading during inclined walking.

Notably, differences were observed between the simulation and experimental profiles, particularly during late stance, where the simulation predicted higher hamstrings activation and KCF peaks than those measured experimentally, and during early stance, where simulated KCF peaks were lower than the experimental observations. These discrepancies are likely attributable to gait adaptations in response to physical assistance, which are not fully captured by the simulation framework that assumes consistent kinematics and idealized force application. Despite these differences in magnitude, the temporal patterns and directional effects of assistance remained consistent across simulation and experiment, supporting the validity of the optimization framework for guiding exoskeleton parameter design.

Overall, these results validate the proposed biomechanics-driven optimization framework. By explicitly incorporating knee contact force and wearability-related parameters into the objective function, the framework identifies exoskeleton configurations that effectively reduce internal knee loading while remaining mechanically feasible and comfortable. The experimental findings further demonstrate that optimizing exoskeleton configuration and control parameters for a specific biomechanical objective can yield measurable benefits beyond those achievable with generic assistance strategies.

Several limitations of this study should be acknowledged. First, the single-subject design limits the generalization of quantitative findings, though it ensured methodological consistency for this proof-of-concept validation. Subjects with different anthropometric or mass properties would alter the feasible parameter bounds and muscle force distributions, likely yielding different optimal solutions. Second, while the optimized assistance was compared against a no-assistance baseline, a direct experimental comparison between pre-optimized and optimized configurations was not conducted. Third, the musculoskeletal model assumed an idealized, massless exoskeleton. Neglecting device inertia and soft-tissue deformation simplifies the optimization but may overlook secondary effects on gait dynamics. Future work will involve larger cohorts and incorporate complex hardware dynamics into the simulation.

V. Conclusion

This study presented a biomechanics-driven framework for optimizing knee exoskeleton configuration and control to reduce internal joint loading during inclined walking. By coupling subject-specific musculoskeletal simulation with PSO, we demonstrated that joint optimization of the exoskeleton configuration and the control strategy effectively modulates muscle recruitment and KCF. Simulation analysis revealed that optimized parameters successfully balance KCF reduction with wearability constraints. Experimental results confirmed substantial reductions in biological knee torque, muscle activations, and KCF, validating the framework's effectiveness in customizing wearable robots for specific biomechanical targets. This work provides a systematic methodology for customizing wearable robots based on internal biomechanical targets. Future research will extend this framework to a larger patient cohort and incorporate more complex device dynamics to enhance the clinical applicability of optimized exoskeleton assistance for joint unloading.

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